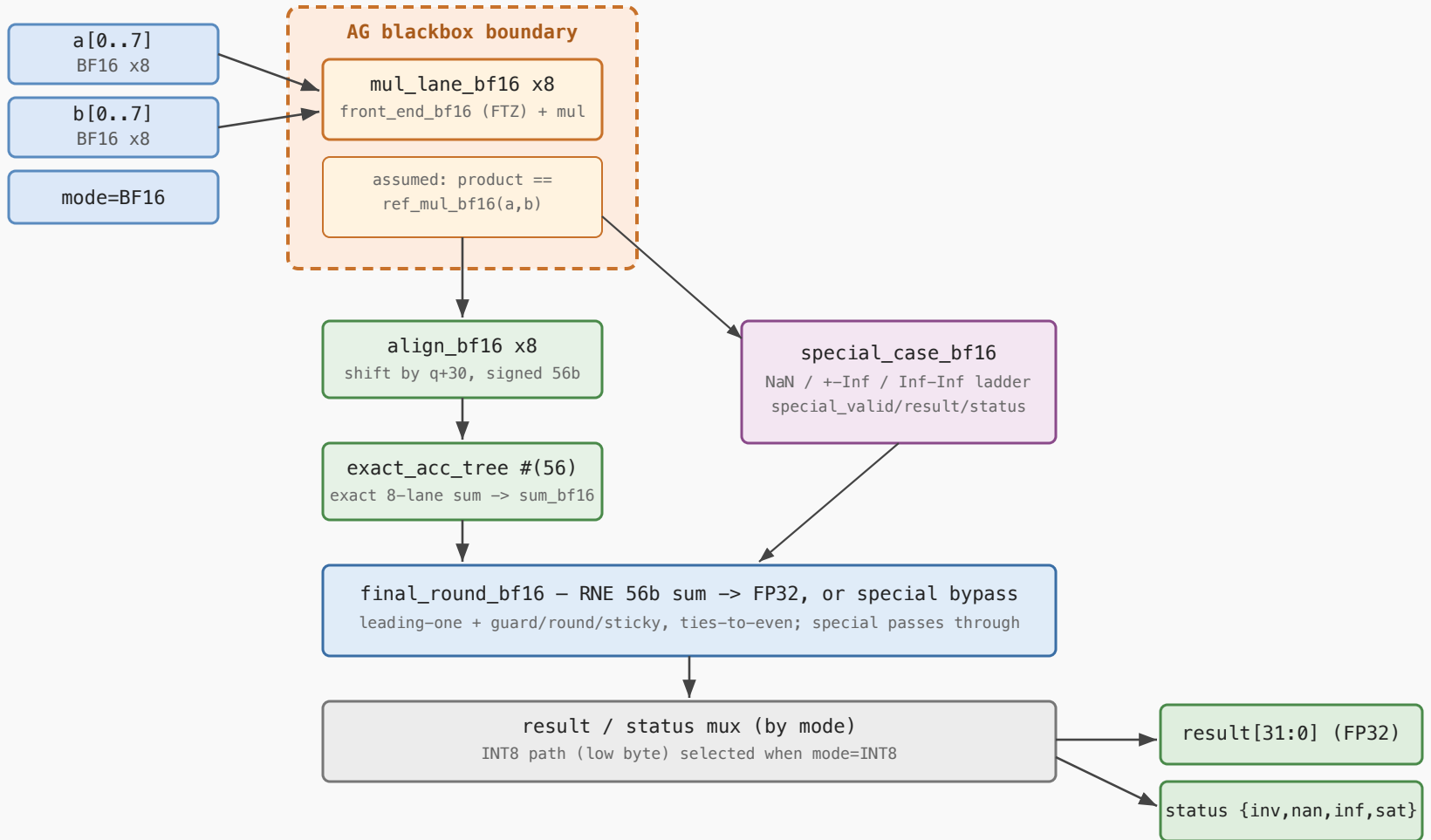


dotprod_top – BF16 Datapath (M3)

N_LANES=8, BF16_W=16, ACC_BF16_W=56 (LSB 2^{30}), FP32_W=32, window E in [119,134]



Assume-Guarantee top proof (four-way, composed by transitivity)

- lane: `mul_lane_bf16 == ref_mul_bf16` (blackboxed above, assumed here)
- align: `align_bf16 lane == ref_align_bf16_lane` (standalone, 8/8 proven)
- round: `final_round_bf16 == ref_round / bypass` (standalone, proven)
- top: `sum_bf16 == sum(aligned lanes) + special ladder == golden pre-round`
 $\Rightarrow$ `result == dotprod_ref_bf16` by transitivity (shifter + rounder kept out of the top miter)